Lunar Base Design Proposal

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Introduction

As the plans for an extensive and possible permanent settlement on the Moon are closer to being executed, it is more important than ever to carefully plan the measures the inhabitants and team on Earth must take to manage technology, architecture, as well as keep track of time as some of the most important considerations. Above all, the team should follow a concise guide, created through examination of previous space missions' achievements and shortcomings. In this way, this proposal will analyze different aspects of a lunar base and propose the most optimal approach to achieving longevity, security, and comfort for inhabitants of lunar settlements. This proposal will also consider the implementation of possible technologies not yet in use, while maintaining relevance to the current technological stage of the scientific community.

Mission Architecture

Firstly, the purpose of establishing an effective and long-lasting lunar base is for a multitude of reasons. A team of scholars specialized in space exploration at the Skolkovo Institute of Science and Technology explain that after the eventual decommission of the ISS, the space industry will focus on integrating into the Lunar system as a way of conducting scientific research and tests, as well as testing new space travel technology that will ultimately help researchers back on Earth innovate and develop more reliable technologies (García, 2016). The result of this is ultimately beneficial to the entire industry by raising the standard of space travel technology and eventually leading humanity to explore farther reaches of space, in parallel, teaching us more about Earth.

Furthermore, in an academic journal by professors of Aerospace at the Technological Institute of Berlin states that the development of the core of a lunar base will take 5 years. After, as the base is being modified, there will be an additional 120 crew members who will join the starting 60 inhabitants of the base over a span of 20 years through separate journeys (Johenning, 1986). As for when to begin the flight to the Moon, NASA and other national space agencies have the technology developed to commence the mission. However, it would be most educated to commence when the crew has been fully prepared and conditioned to avoid the jeopardization of the mission. Conclusively, the arrival of the mission depends on the preparation of the crew and the on-ground team; however, there is a timeline that estimates the duration between each event after the crew has arrived on board.

Base Goals and Expected Outcomes

A primary motivator for the establishment of a lunar base would be to run environmental tests and conduct research on the natural aspects of the Moon. The journal developed by aerospace scholars at the Skolkovo Institute of Technology explains that scientists will be able to research the effects of lunar radiation on humans and help us improve living conditions for inhabitants of the moon (García, 2016). This will also benefit the larger population of humans on Earth by highlighting aspects of how human bodies react to radiation in a much clearer manner. The journal also describes how establishing the base would also assist in creating an environment for astronauts to undergo test missions that will prepare them for future missions to Mars. This way, the base would both operate as a facility to learn more about celestial bodies and explain human nature, as well as promote space exploration to farther regions and other celestial bodies.

Base Location and Structure

A crucial step in planning a Lunar base is determining the ideal location and structural build to accommodate the base’s purposes. An article published by a scientific journalist at NewScientist explains that due to the plentiful water source detected in the vicinity of two craters at the south pole of the Moon suggests it to be the most ideal location for a base (Wilkins, 2013). The creation of a base in this region would be ideal for supplying natural materials, mainly water, to research organic matter, as well s supply drinking water for humans. One of the most sought-after topics in research, which intrigues scientists, is the behavior of plants in space; for this reason, a plentiful water supply would decrease the need for additional missions to bring water to the base researchers.

Base Governance

The governance of the lunar base should be a partnership between all the nations contributing to the missions and the base as a whole. Similar to the International Space Station, anyone contributing to the project as a whole should be able to have researchers and astronauts from the country conduct research and practice their individuality on the lunar base. An article published by a science journalist at the BBC explains that a lunar base should not belong to an individual country or organization, as this would contradict the principles of the Outer Space Treaty of 1967, which states that no one can claim ownership of the Moon (Morelle, 2024). There should also be some limits set on the interactions between the countries involved in the lunar base, as the article explains how international rivalries could be transferred to the individuals on the base and cause internal conflict, ultimately detrimental to the missions.

Base Operational Concepts

A lunar base would be most effective if it were accompanied by operational technologies that would assist in the inhabitants' missions. An article published by NASA explaining potential aiding tools for the second venture to the moon emphasizes the use of a lunar terrain vehicle (LTV) (Warner, 2020). A vehicle would be beneficial to the inhabitants of the lunar base as it would allow for easier transportation across craters and help traverse the rocky and challenging terrain that the moon is known for. The article also suggests the use of commercial lunar aid services, in which NASA would partner with companies in order to make supply trips of tools to aid the moon's inhabitants. Secure transportation of supplies would be crucial in maintaining longevity for the lunar outpost. It would guarantee dates for when materials and commodities would be shipped to the base to aid astronauts in both living as well as research. Furthermore, NASA's article also proposes the use of radiation shields to limit the amount of radiation endured by the inhabitants of the lunar settlements. This would greatly improve the health and safety of the astronauts by limiting health problems, including the risk of cancer. This would ultimately allow for more thorough research and increased missions as it would allow astronauts to stay on the lunar surface and interact with the lunar base for a longer period of time.

Conclusion

To conclude, a lunar base should be equipped with essential operational technologies to aid the astronauts in their mission while also having an authority structure. By implementing a partnership role of governance for the lunar base, it will allow for more diversification in technological development and more contributors to the expansion of the base. This same process is reflected in the International Space Station, in which collaboration enhanced the workflow and worked as a symbol of alliance between the nations, especially in the case of the

United States and Russia following the Cold War. In total, the implementation of a lunar base would help teach scientists more about space, as well as Earth, through investigation of the materials and matter on the Moon, while existing as a symbol of unity that breaks the national boundaries that have been placed through long periods of conflict on Earth.

References

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